

Comp Prog Code Compilation

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BSCS - 1 (CBM)

Midterm

Comp Prog

Drill 2 - Accept Two Values and Display the Sum, Difference, Quotient and Product

<pre>1 #include <iostream> 2 using namespace std; 3 4 int main() { 5 double num1, num2; 6 7 cout << "Enter the first number: "; 8 cin >> num1; 9 cout << "Enter the second number: "; 10 cin >> num2; 11 12 double sum = num1 + num2; 13 double difference = num1 - num2; 14 double product = num1 * num2; 15 16 if (num2 != 0) { 17 double quotient = num1 / num2; 18 cout << "Quotient: " << quotient << endl; 19 } else { 20 cout << "Cannot divide by zero!" << endl; 21 } 22 23 cout << "Sum: " << sum << endl; 24 cout << "Difference: " << difference << endl; 25 cout << "Product: " << product << endl; 26 27 return 0; 28 }</pre>	<pre>Enter the first number: 10 Enter the second number: 7 Quotient: 1.42857 Sum: 17 Difference: 3 Product: 70 === Code Execution Successful ===</pre>
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Drill 3 - Odd or Even Value

<pre>1 #include <iostream> 2 using namespace std; 3 4 int main() { 5 int num; 6 7 cout << "Enter an integer: "; 8 cin >> num; 9 10 if (num % 2 == 0) { 11 cout << num << " is even." << endl; 12 } else { 13 cout << num << " is odd." << endl; 14 } 15 16 return 0; 17 }</pre>	<pre>Enter an integer: 5 5 is odd. === Code Execution Successful ===</pre>
<pre>1 #include <iostream> 2 using namespace std; 3 4 int main() { 5 int num; 6 7 cout << "Enter an integer: "; 8 cin >> num; 9 10 if (num % 2 == 0) { 11 cout << num << " is even." << endl; 12 } else { 13 cout << num << " is odd." << endl; 14 } 15 16 return 0; 17 }</pre>	<pre>Enter an integer: 4 4 is even. === Code Execution Successful ===</pre>

Feb 21, 2025. Drill 2 - Accept Two Values and Display the Sum, Difference, Quotient and Product

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     double num1, num2;
6
7     cout << "Enter the first number: ";
8     cin >> num1;
9     cout << "Enter the second number: ";
10    cin >> num2;
11
12    double sum = num1 + num2;
13    double difference = num1 - num2;
14    double product = num1 * num2;
15
16    if (num2 != 0) {
17        double quotient = num1 / num2;
18        cout << "Quotient: " << quotient << endl;
19    } else {
20        cout << "Cannot divide by zero!" << endl;
21    }
22
23    cout << "Sum: " << sum << endl;
24    cout << "Difference: " << difference << endl;
25    cout << "Product: " << product << endl;
26
27    return 0;
28 }
```

Enter the first number: 0
Enter the second number: 0
Cannot divide by zero!
Sum: 0
Difference: 0
Product: 0

=== Code Execution Successful ===

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     double num1, num2;
6
7     cout << "Enter the first number: ";
8     cin >> num1;
9     cout << "Enter the second number: ";
10    cin >> num2;
11
12    double sum = num1 + num2;
13    double difference = num1 - num2;
14    double product = num1 * num2;
15
16    if (num2 != 0) {
17        double quotient = num1 / num2;
18        cout << "Quotient: " << quotient << endl;
19    } else {
20        cout << "Cannot divide by zero!" << endl;
21    }
22
23    cout << "Sum: " << sum << endl;
24    cout << "Difference: " << difference << endl;
25    cout << "Product: " << product << endl;
26
27    return 0;
28 }
```

Enter the first number: 100
Enter the second number: 10
Quotient: 10
Sum: 110
Difference: 90
Product: 1000

=== Code Execution Successful ===

Feb. 21, 2025. Drill 4 - Prelims, Midterm, Finals, and Semestral Grade

<pre>1 #include <iostream> 2 using namespace std; 3 4 int main() { 5 // Declare variables to store grades 6 float prelim, midterm, final, semestral_grade; 7 8 // Get input for each grade 9 cout << "Enter Prelim Grade: "; 10 cin >> prelim; 11 12 cout << "Enter Midterm Grade: "; 13 cin >> midterm; 14 15 cout << "Enter Final Grade: "; 16 cin >> final; 17 18 // Calculate the semestral grade using the given weights 19 semestral_grade = (prelim * 0.30) + (midterm * 0.30) + (final * 0.40); 20 21 // Display the semestral grade 22 cout << "The Semestral Grade is: " << semestral_grade << endl; 23 24 return 0; 25 }</pre>	<pre>Enter Prelim Grade: 91 Enter Midterm Grade: 94 Enter Final Grade: 86 The Semestral Grade is: 89.9 === Code Execution Successful ===</pre>
---	---

Feb. 21, 2025. Drill 5 - Grade < 75 Passed, if else Failed

<pre>1 #include <iostream> 2 using namespace std; 3 4 int main() { 5 float grade; 6 7 cout << "Enter the grade: "; 8 cin >> grade; 9 10 if (grade >= 75) { 11 cout << "Passed" << endl; 12 } else { 13 cout << "Failed" << endl; 14 } 15 16 return 0; 17 }</pre>	<pre>Enter the grade: 75 Passed === Code Execution Successful ===</pre>
---	--

<pre>1 #include <iostream> main.cpp 2 using namespace std; 3 4 int main() { 5 float grade; 6 7 cout << "Enter the grade: "; 8 cin >> grade; 9 10 if (grade >= 75) { 11 cout << "Passed" << endl; 12 } else { 13 cout << "Failed" << endl; 14 } 15 16 return 0; 17 }</pre>	<pre>Enter the grade: 70 Failed === Code Execution Successful ===</pre>
--	--

Feb 21, 2025. Name, Age, Address, Contact number, Course, Year

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     // Declare variables
6     char name[100];
7     int age;
8     char address[200];
9     char contactNumber[15];
10    char course[50];
11    int year;
12
13    // Accept user input
14    cout << "Enter your name: ";
15    cin.getline(name, sizeof(name));
16
17    cout << "Enter your age: ";
18    cin >> age;
19    cin.ignore(); // To ignore the newline character left by the previous input
20
21    cout << "Enter your address: ";
22    cin.getline(address, sizeof(address));
23
24    cout << "Enter your contact number: ";
25    cin.getline(contactNumber, sizeof(contactNumber));
26
27    cout << "Enter your course: ";
28    cin.getline(course, sizeof(course));
29
30    cout << "Enter your year: ";
31    cin >> year;
32
33    // Display the entered information
34    cout << "\n--- Displaying Entered Information ---\n";
35    cout << "Name: " << name << endl;
36    cout << "Age: " << age << endl;
37    cout << "Address: " << address << endl;
38    cout << "Contact Number: " << contactNumber << endl;
39    cout << "Course: " << course << endl;
40    cout << "Year: " << year << endl;
41
42    return 0;
43 }
```

Enter your name: Joeffrey Cadorna
Enter your age: 19
Enter your address: Pandacan, Manila
Enter your contact number: 09154254399
Enter your course: BSCS
Enter your year: 1

--- Displaying Entered Information ---
Name: Joeffrey Cadorna
Age: 19
Address: Pandacan, Manila
Contact Number: 09154254399
Course: BSCS
Year: 1

=== Code Execution Successful ===

Feb 21, 2025. Employee name, rate, and salary

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      // Declare variables
6      char employeeName[100];
7      float ratePerHour;
8      float hoursWorked;
9      float deduction;
10     float salary;
11
12     // Accept user input
13     cout << "Enter Employee Name: ";
14     cin.getline(employeeName, sizeof(employeeName));
15
16     cout << "Enter Rate per Hour: $";
17     cin >> ratePerHour;
18
19     cout << "Enter Hours Worked: ";
20     cin >> hoursWorked;
21
22     cout << "Enter Deduction: $";
23     cin >> deduction;
24
25     // Calculate salary
26     salary = (ratePerHour * hoursWorked) - deduction;
27
28     // Display the salary
29     cout << "\n--- Salary Details ---\n";
30     cout << "Employee Name: " << employeeName << endl;
31     cout << "Rate per Hour: $" << ratePerHour << endl;
32     cout << "Hours Worked: " << hoursWorked << endl;
33     cout << "Deduction: $" << deduction << endl;
34     cout << "Salary: $" << salary << endl;
35
36     return 0;
37 }
```

```
Enter Employee Name: Aristo Tel
Enter Rate per Hour: $12
Enter Hours Worked: 79
Enter Deduction: $33
```

```
--- Salary Details ---
Employee Name: Aristo Tel
Rate per Hour: $12
Hours Worked: 79
Deduction: $33
Salary: $915
```

```
=== Code Execution Successful ===
```

Feb. 28, 2025. Quiz - Product name, No. of orders, Amount per pc, and Total amount to pay

<pre>1 #include <iostream> 2 #include <string> 3 using namespace std; 4 5 int main() { 6 string productName; 7 int numOrders; 8 double amountPerPiece, totalAmount; 9 10 cout << "Enter product name: "; 11 cin >> productName; 12 13 cout << "Enter number of orders: "; 14 cin >> numOrders; 15 16 cout << "Enter amount per piece: \$"; 17 cin >> amountPerPiece; 18 19 totalAmount = numOrders * amountPerPiece; 20 21 cout << "\nProduct: " << productName << endl; 22 cout << "Number of orders: " << numOrders << endl; 23 cout << "Amount per piece: \$" << amountPerPiece << endl; 24 cout << "Total amount to pay: \$" << totalAmount << endl; 25 26 return 0; 27 }</pre>	<pre>Enter product name: Mouse Enter number of orders: 5 Enter amount per piece: \$17 Product: Mouse Number of orders: 5 Amount per piece: \$17 Total amount to pay: \$85 === Code Execution Successful ===</pre>
---	---

March 07, 2025. Activity 1 - Student Name, Prelim, Midterm, Final, and Semestral Grade

```
1  #include <iostream>
2  #include <string>
3  using namespace std;
4
5  int main() {
6      string studentName;
7      float prelimGrade, midtermGrade, finalGrade, semestralGrade;
8
9      // Input student details
10     cout << "Enter student name: ";
11     getline(cin, studentName);
12
13     cout << "Enter Prelim grade: ";
14     cin >> prelimGrade;
15
16     cout << "Enter Midterm grade: ";
17     cin >> midtermGrade;
18
19     cout << "Enter Final grade: ";
20     cin >> finalGrade;
21
22     // Calculate the semestral grade based on the given formula
23     semestralGrade = (prelimGrade * 0.30) + (midtermGrade * 0.30) + (finalGrade * 0.40);
24
25     // Output the semestral grade
26     cout << "\nStudent Name: " << studentName << endl;
27     cout << "Prelim Grade: " << prelimGrade << endl;
28     cout << "Midterm Grade: " << midtermGrade << endl;
29     cout << "Final Grade: " << finalGrade << endl;
30     cout << "Semestral Grade: " << semestralGrade << endl;
31
32     return 0;
33 }
```

Enter student name: Socra Tes
Enter Prelim grade: 97
Enter Midterm grade: 98
Enter Final grade: 96

Student Name: Socra Tes
Prelim Grade: 97
Midterm Grade: 98
Final Grade: 96
Semestral Grade: 96.9

=== Code Execution Successful ===

March 07, 2025. Activity 2 - Product name, No. of orders, Amount per pc, VAT and Total amount to pay

<pre>1 #include <iostream> 2 #include <string> 3 using namespace std; 4 5 int main() { 6 string productName; 7 int numberOfOrders; 8 float pricePerPiece, totalAmount, vat, amountToPay; 9 10 // Input the product details 11 cout << "Enter product name: "; 12 getline(cin, productName); // To accept a string with spaces 13 14 cout << "Enter number of orders: "; 15 cin >> numberOfOrders; 16 17 cout << "Enter price per piece: "; 18 cin >> pricePerPiece; 19 20 // Calculate the total amount to pay (number of orders * price per piece) 21 totalAmount = numberOfOrders * pricePerPiece; 22 23 // Calculate VAT (12% of the total amount) 24 vat = totalAmount * 0.12; 25 26 // Calculate the final amount to pay (total amount + VAT) 27 amountToPay = totalAmount + vat; 28 29 // Output the result 30 cout << "\nProduct Name: " << productName << endl; 31 cout << "Number of Orders: " << numberOfOrders << endl; 32 cout << "Price per Piece: " << pricePerPiece << endl; 33 cout << "Total Amount: " << totalAmount << endl; 34 cout << "VAT (12%): " << vat << endl; 35 cout << "Amount to Pay (Total + VAT): " << amountToPay << endl; 36 37 return 0; 38 }</pre>	<pre>Enter product name: Lamp Enter number of orders: 7 Enter price per piece: 13 Product Name: Lamp Number of Orders: 7 Price per Piece: 13 Total Amount: 91 VAT (12%): 10.92 Amount to Pay (Total + VAT): 101.92 === Code Execution Successful ===</pre>
---	--

DBMS

Drill 1 - MySQL Student Name, Age, Course, Year, and Department

```
1 -- create
2 CREATE TABLE STUDENT (
3     studentId INTEGER PRIMARY KEY,
4     name TEXT NOT NULL,
5     age INTEGER NOT NULL,
6     course TEXT NOT NULL,
7     year INTEGER NOT NULL,
8     department TEXT NOT NULL
9 );
10
11 -- insert example data
12 INSERT INTO STUDENT VALUES (1, 'Ryan Felipe', 20, 'Computer Science', 2, 'CBM');
13 INSERT INTO STUDENT VALUES (2, 'Julius Rodriguez', 21, 'Information Technology', 3, 'CBM');
14 INSERT INTO STUDENT VALUES (3, 'Roberto Penano', 22, 'Entrepreneurship', 4, 'CBM');
15 INSERT INTO STUDENT VALUES (4, 'Allysa Jane Cruz', 19, 'Computer Science', 1, 'CBM');
16 INSERT INTO STUDENT VALUES (5, 'Zachary Flores', 19, 'Tourism Management', 1, 'CBM');
17 INSERT INTO STUDENT VALUES (6, 'Ashley Gabriel Reyes', 20, 'Hospitality Management', 2, 'CBM');
18 INSERT INTO STUDENT VALUES (7, 'Isabella Martinzel', 21, 'Nursing', 3, 'CNAHS');
19 INSERT INTO STUDENT VALUES (8, 'Reynald Bautista', 19, 'Music Therapy', 1, 'CMPA');
20 INSERT INTO STUDENT VALUES (9, 'Hanna Garcia', 20, 'Tourism Management', 4, 'CBM');
21 INSERT INTO STUDENT VALUES (10, 'Claire Aquino', 19, 'Computer Science', 1, 'CBM');
22
23 -- fetch all students
24 SELECT * FROM STUDENT;
```

STDIN

Input for the program (Optional)

Output:

studentId	name
1	Ryan Felipe
2	Julius Rodriguez
3	Roberto Penano
4	Allysa Jane Cruz
5	Zachary Flores
6	Ashley Gabriel Reyes
7	Isabella Martinzel
8	Reynald Bautista
9	Hanna Garcia
10	Claire Aquino

```
1 -- create
2 CREATE TABLE STUDENT (
3     studentId INTEGER PRIMARY KEY,
4     name TEXT NOT NULL,
5     age INTEGER NOT NULL,
6     course TEXT NOT NULL,
7     year INTEGER NOT NULL,
8     department TEXT NOT NULL
9 );
10
11 -- insert example data
12 INSERT INTO STUDENT VALUES (1, 'Ryan Felipe', 20, 'Computer Science', 2, 'CBM');
13 INSERT INTO STUDENT VALUES (2, 'Julius Rodriguez', 21, 'Information Technology', 3, 'CBM');
14 INSERT INTO STUDENT VALUES (3, 'Roberto Penano', 22, 'Entrepreneurship', 4, 'CBM');
15 INSERT INTO STUDENT VALUES (4, 'Allysa Jane Cruz', 19, 'Computer Science', 1, 'CBM');
16 INSERT INTO STUDENT VALUES (5, 'Zachary Flores', 19, 'Tourism Management', 1, 'CBM');
17 INSERT INTO STUDENT VALUES (6, 'Ashley Gabriel Reyes', 20, 'Hospitality Management', 2, 'CBM');
18 INSERT INTO STUDENT VALUES (7, 'Isabella Martinzel', 21, 'Nursing', 3, 'CNAHS');
19 INSERT INTO STUDENT VALUES (8, 'Reynald Bautista', 19, 'Music Therapy', 1, 'CMPA');
20 INSERT INTO STUDENT VALUES (9, 'Hanna Garcia', 20, 'Tourism Management', 4, 'CBM');
21 INSERT INTO STUDENT VALUES (10, 'Claire Aquino', 19, 'Computer Science', 1, 'CBM');
22
23 -- fetch all students
24 SELECT * FROM STUDENT;
```

STDIN

Input for the program (Optional)

age	course
20	Computer Science
21	Information Technology
22	Entrepreneurship
19	Computer Science
19	Tourism Management
20	Hospitality Management
21	Nursing
19	Music Therapy
20	Tourism Management
19	Computer Science

```
1 -- create
2 CREATE TABLE STUDENT (
3     studentId INTEGER PRIMARY KEY,
4     name TEXT NOT NULL,
5     age INTEGER NOT NULL,
6     course TEXT NOT NULL,
7     year INTEGER NOT NULL,
8     department TEXT NOT NULL
9 );
10
11 -- insert example data
12 INSERT INTO STUDENT VALUES (1, 'Ryan Felipe', 20, 'Computer Science', 2, 'CBM');
13 INSERT INTO STUDENT VALUES (2, 'Julius Rodriguez', 21, 'Information Technology', 3, 'CBM');
14 INSERT INTO STUDENT VALUES (3, 'Roberto Penano', 22, 'Entrepreneurship', 4, 'CBM');
15 INSERT INTO STUDENT VALUES (4, 'Allysa Jane Cruz', 19, 'Computer Science', 1, 'CBM');
16 INSERT INTO STUDENT VALUES (5, 'Zachary Flores', 19, 'Tourism Management', 1, 'CBM');
17 INSERT INTO STUDENT VALUES (6, 'Ashley Gabriel Reyes', 20, 'Hospitality Management', 2, 'CBM');
18 INSERT INTO STUDENT VALUES (7, 'Isabella Martinzel', 21, 'Nursing', 3, 'CNAHS');
19 INSERT INTO STUDENT VALUES (8, 'Reynald Bautista', 19, 'Music Therapy', 1, 'CMPA');
20 INSERT INTO STUDENT VALUES (9, 'Hanna Garcia', 20, 'Tourism Management', 4, 'CBM');
21 INSERT INTO STUDENT VALUES (10, 'Claire Aquino', 19, 'Computer Science', 1, 'CBM');
22
23 -- fetch all students
24 SELECT * FROM STUDENT;
```

STDIN

Input for the program (Optiona

year	department
2	CBM
3	CBM
4	CBM
1	CBM
1	CBM
2	CBM
3	CNAHS
1	CMPA
4	CBM
1	CBM

Feb 21, 2025. SQL Code

```
CREATE TABLE customer (
  customer_id INT IDENTITY(1,1) PRIMARY KEY,
  first_name VARCHAR(50) NOT NULL,
  last_name VARCHAR(50) NOT NULL,
  email VARCHAR(100) NOT NULL UNIQUE,
  phone_number VARCHAR(15),
  address VARCHAR(255)
);

INSERT INTO customer (first_name, last_name, email, phone_number, address) VALUES
('allysa jane', 'dela cruz', 'allygmail.com', '003-567', 'zamora manila'),
('jasmine marie', 'jordan', 'jmariejgmail.com', '023-597', 'pandacan manila'),
('myca louise', 'ang', 'mycalouiseanggmail.com', '343-5127', 'paco manila'),
('sophia beatrice', 'mendoza', 'piazyygmail.com', '453-5237', 'sta ana manila'),
('paul dominique', 'valdevieso', 'pauldvgmail.com', '893-51267', 'sta ana manila'),
('justin', 'abarico', 'tinabaricogmail.com', '777-067', 'sta mesa manila'),
('yuna', 'reyes', 'yunayigmail.com', '233-04807', 'intramuros manila'),
('allison', 'myoui', 'allymyouigmail.com', '753-6517', 'san andres manila'),
('jelo', 'delos angeles', 'jeldagmail.com', '453-21312', 'quiapo manila'),
('katarina', 'medina', 'katarinamdgmail.com', '893-52337', 'tondo manila');

SELECT first_name, last_name, email, phone_number, address FROM customer;
```

STDIN

Input for the program (Optional)

Output:

first_name	last_name	email
allysa jane	dela cruz	allygmail.com
jasmine marie	jordan	jmariejgmail.com
myca louise	ang	mycalouiseanggmail.com
sophia beatrice	mendoza	piazyygmail.com
paul dominique	valdevieso	pauldvgmail.com
justin	abarico	tinabaricogmail.com
yuna	reyes	yunayigmail.com
allison	myoui	allymyouigmail.com
jelo	delos angeles	jeldagmail.com
katarina	medina	katarinamdemail.com

```
CREATE TABLE customer (
  customer_id INT IDENTITY(1,1) PRIMARY KEY,
  first_name VARCHAR(50) NOT NULL,
  last_name VARCHAR(50) NOT NULL,
  email VARCHAR(100) NOT NULL UNIQUE,
  phone_number VARCHAR(15),
  address VARCHAR(255)
);

INSERT INTO customer (first_name, last_name, email, phone_number, address) VALUES
('allysa jane', 'dela cruz', 'allygmail.com', '003-567', 'zamora manila'),
('jasmine marie', 'jordan', 'jmariejgmail.com', '023-597', 'pandacan manila'),
('myca louise', 'ang', 'mycalouiseanggmail.com', '343-5127', 'paco manila'),
('sophia beatrice', 'mendoza', 'piazyygmail.com', '453-5237', 'sta ana manila'),
('paul dominique', 'valdevieso', 'pauldvgmail.com', '893-51267', 'sta ana manila'),
('justin', 'abarico', 'tinabaricogmail.com', '777-067', 'sta mesa manila'),
('yuna', 'reyes', 'yunayigmail.com', '233-04807', 'intramuros manila'),
('allison', 'myoui', 'allymyouigmail.com', '753-6517', 'san andres manila'),
('jelo', 'delos angeles', 'jeldagmail.com', '453-21312', 'quiapo manila'),
('katarina', 'medina', 'katarinamdgmail.com', '893-52337', 'tondo manila');

SELECT first_name, last_name, email, phone_number, address FROM customer;
```

STDIN

Input for the program (Optional)

phone_number	address
003-567	zamora manila
023-597	pandacan manila
343-5127	paco manila
453-5237	sta ana manila
893-51267	sta ana manila
777-067	sta mesa manila
233-04807	intramuros manila
753-6517	san andres manila
453-21312	quiapo manila
893-52337	tondo manila

Pre-finals

Comp Prog

March 26, 2025. Assignment - If Else If

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int grade;
6
7     // Ask user to enter the grade
8     cout << "Enter the grade: ";
9     cin >> grade;
10
11    // Determine and display the message based on the grade
12    if (grade <= 70) {
13        cout << "Oh no Failed!" << endl;
14    } else if (grade <= 75) {
15        cout << "Study harder" << endl;
16    } else {
17        cout << "Passed" << endl;
18    }
19
20    return 0;
21 }
```

Enter the grade: 70

Oh no Failed!

=== Code Execution Successful ===

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int grade;
6
7     // Ask user to enter the grade
8     cout << "Enter the grade: ";
9     cin >> grade;
10
11    // Determine and display the message based on the grade
12    if (grade <= 70) {
13        cout << "Oh no Failed!" << endl;
14    } else if (grade <= 75) {
15        cout << "Study harder" << endl;
16    } else {
17        cout << "Passed" << endl;
18    }
19
20    return 0;
21 }
22
```

Enter the grade: 75

Study harder

=== Code Execution Successful ===

<pre> 1 #include <iostream> 2 using namespace std; 3 4 int main() { 5 int grade; 6 7 // Ask user to enter the grade 8 cout << "Enter the grade: "; 9 cin >> grade; 10 11 // Determine and display the message based on the grade 12 if (grade <= 70) { 13 cout << "Oh no Failed!" << endl; 14 } else if (grade <= 75) { 15 cout << "Study harder" << endl; 16 } else { 17 cout << "Passed" << endl; 18 } 19 20 return 0; 21 }</pre>	<pre> Enter the grade: 99 Passed === Code Execution Successful ===</pre>
--	---

March 28, 2025. Activity 1 - Given code, to be run in programiz and analyzed in ½ crosswise)

<pre> 1 #include <iostream> 2 using namespace std; 3 int main() { 4 5 double x, y; 6 cout << "\nEnter two floating-point values: "; 7 cin >> x >> y; 8 cout << "The average of the two number is: " 9 << (x+y)/2.0 << endl; 10 11 return 0; 12 }</pre>	<pre> Enter two floating-point values: 13.9 33.7 The average of the two number is: 23.8 === Code Execution Successful ===</pre>
---	--

March 28, 2025. Activity 2 - Given code, to be run in programiz and analyzed in ½ crosswise)

<pre> 1 #include <iostream> 2 using namespace std; 3 int main() 4 { 5 int i(2), j(8); 6 7 cout <<i++ <<endl; 8 cout <<i <<endl; 9 cout <<j-- <<endl; 10 cout <<--j <<endl; 11 12 return 0; 13 }</pre>	<pre> 2 3 8 6 === Code Execution Successful ===</pre>
---	--

March 28, 2025. Activity 3 - Given code, to be run in programiz and analyzed in ½ crosswise)

<pre>1 // Demonstration of compound assignments 2 3 #include <iostream> 4 #include <iomanip> 5 using namespace std; 6 7 int main () 8 { 9 float x, y; 10 11 cout << "\n Please enter a starting value: "; 12 cin >> x; 13 14 cout << "\n Please enter the increment value: "; 15 cin >> y; 16 17 x += y; 18 19 cout << "\n And now multiplication! "; 20 cout << "\n Please enter a factor: "; 21 cin >> y; 22 23 x *= y; 24 25 cout << "\n Finally division."; 26 cout << "\n Please supply a divisor: "; 27 cin >> y; 28 29 x /= y; 30 31 cout << "\n And this is " 32 << "your current lucky number: " 33 << " // without digits after 34 << " // the decimal point: 35 << fixed << setprecision(0) 36 << x << endl; 37 return 0; 38 }</pre>	<pre>Please enter a starting value: 1 Please enter the increment value: 2 And now multiplication! Please enter a factor: 2 Finally division. Please supply a divisor: 4 And this is your current lucky number: 2 === Code Execution Successful ===</pre>
---	---

April 11, 2025. Drill 1 - if-else if-else that will accept the general average & determine what strand does it belong. General Average of 75 to 80 = HUMMS, 81 to 85 = ABM. 86 to 100 = STEM. Otherwise "see college department for information"

```

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      float generalAverage;
6
7      // Ask for user input
8      cout << "Enter your General Average: ";
9      cin >> generalAverage;
10
11     // Determine strand based on general average
12     if (generalAverage >= 75 && generalAverage <= 80) {
13         cout << "You belong to the HUMMS strand." << endl;
14     }
15     else if (generalAverage >= 81 && generalAverage <= 85) {
16         cout << "You belong to the ABM strand." << endl;
17     }
18     else if (generalAverage >= 86 && generalAverage <= 100) {
19         cout << "You belong to the STEM strand." << endl;
20     }
21     else {
22         cout << "Please see the college department for more information." <<
            endl;
23     }
24
25     return 0;
26 }

```

Enter your General Average: 78
You belong to the HUMMS strand.

=== Code Execution Successful ===

```

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      float generalAverage;
6
7      // Ask for user input
8      cout << "Enter your General Average: ";
9      cin >> generalAverage;
10
11     // Determine strand based on general average
12     if (generalAverage >= 75 && generalAverage <= 80) {
13         cout << "You belong to the HUMMS strand." << endl;
14     }
15     else if (generalAverage >= 81 && generalAverage <= 85) {
16         cout << "You belong to the ABM strand." << endl;
17     }
18     else if (generalAverage >= 86 && generalAverage <= 100) {
19         cout << "You belong to the STEM strand." << endl;
20     }
21     else {
22         cout << "Please see the college department for more information." <<
            endl;
23     }
24
25     return 0;
26 }

```

Enter your General Average: 84
You belong to the ABM strand.

=== Code Execution Successful ===

```

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      float generalAverage;
6
7      // Ask for user input
8      cout << "Enter your General Average: ";
9      cin >> generalAverage;
10
11     // Determine strand based on general average
12     if (generalAverage >= 75 && generalAverage <= 80) {
13         cout << "You belong to the HUMMS strand." << endl;
14     }
15     else if (generalAverage >= 81 && generalAverage <= 85) {
16         cout << "You belong to the ABM strand." << endl;
17     }
18     else if (generalAverage >= 86 && generalAverage <= 100) {
19         cout << "You belong to the STEM strand." << endl;
20     }
21     else {
22         cout << "Please see the college department for more information." <<
            endl;
23     }
24
25     return 0;
26 }

```

Enter your General Average: 94
You belong to the STEM strand.

=== Code Execution Successful ===

```

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      float generalAverage;
6
7      // Ask for user input
8      cout << "Enter your General Average: ";
9      cin >> generalAverage;
10
11     // Determine strand based on general average
12     if (generalAverage >= 75 && generalAverage <= 80) {
13         cout << "You belong to the HUMMS strand." << endl;
14     }
15     else if (generalAverage >= 81 && generalAverage <= 85) {
16         cout << "You belong to the ABM strand." << endl;
17     }
18     else if (generalAverage >= 86 && generalAverage <= 100) {
19         cout << "You belong to the STEM strand." << endl;
20     }
21     else {
22         cout << "Please see the college department for more information." <<
            endl;
23     }
24
25     return 0;
26 }

```

Enter your General Average: 70
Please see the college department for more information.

=== Code Execution Successful ===

April 11, 2025. Quiz - if-else if-else that will accept user input (grade) & determine if scholarship of sm if grade - if 81-85 = SM Foundation, 86 to 90 = SMDC Scholar, 91 to 95 = NU Scholar, 96 to 100 = Lance Sy Scholar, otherwise "No Scholar Available"

```

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      float generalAverage;
6
7      cout << "Enter your General Average: ";
8      cin >> generalAverage;
9
10     if (generalAverage >= 81 && generalAverage <= 85) {
11         cout << "Applicable for SM Foundation" << endl;
12     }
13     else if (generalAverage >= 86 && generalAverage <= 90) {
14         cout << "Applicable for SMDC Scholar" << endl;
15     }
16     else if (generalAverage >= 91 && generalAverage <= 95) {
17         cout << "Applicable for NU Scholar" << endl;
18     }
19     else if (generalAverage >= 96 && generalAverage <= 100) {
20         cout << "Applicable for Lance Sy Scholar." << endl;
21     }
22     else {
23         cout << "No Scholarship Available." << endl;
24     }
25
26     return 0;
27 }

```

Enter your General Average: 84
Applicable for SM Foundation

=== Code Execution Successful ===

```

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      float generalAverage;
6
7      cout << "Enter your General Average: ";
8      cin >> generalAverage;
9
10     if (generalAverage >= 81 && generalAverage <= 85) {
11         cout << "Applicable for SM Foundation" << endl;
12     }
13     else if (generalAverage >= 86 && generalAverage <= 90) {
14         cout << "Applicable for SMDC Scholar" << endl;
15     }
16     else if (generalAverage >= 91 && generalAverage <= 95) {
17         cout << "Applicable for NU Scholar" << endl;
18     }
19     else if (generalAverage >= 96 && generalAverage <= 100) {
20         cout << "Applicable for Lance Sy Scholar." << endl;
21     }
22     else {
23         cout << "No Scholarship Available." << endl;
24     }
25
26     return 0;
27 }

```

Enter your General Average: 87
Applicable for SMDC Scholar

=== Code Execution Successful ===

```

1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     float generalAverage;
6
7     cout << "Enter your General Average: ";
8     cin >> generalAverage;
9
10    if (generalAverage >= 81 && generalAverage <= 85) {
11        cout << "Applicable for SM Foundation" << endl;
12    }
13    else if (generalAverage >= 86 && generalAverage <= 90) {
14        cout << "Applicable for SMDC Scholar" << endl;
15    }
16    else if (generalAverage >= 91 && generalAverage <= 95) {
17        cout << "Applicable for NU Scholar" << endl;
18    }
19    else if (generalAverage >= 96 && generalAverage <= 100) {
20        cout << "Applicable for Lance Sy Scholar." << endl;
21    }
22    else {
23        cout << "No Scholarship Available." << endl;
24    }
25
26    return 0;
27 }

```

Enter your General Average: 93
Applicable for NU Scholar

=== Code Execution Successful ===

```

1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     float generalAverage;
6
7     cout << "Enter your General Average: ";
8     cin >> generalAverage;
9
10    if (generalAverage >= 81 && generalAverage <= 85) {
11        cout << "Applicable for SM Foundation" << endl;
12    }
13    else if (generalAverage >= 86 && generalAverage <= 90) {
14        cout << "Applicable for SMDC Scholar" << endl;
15    }
16    else if (generalAverage >= 91 && generalAverage <= 95) {
17        cout << "Applicable for NU Scholar" << endl;
18    }
19    else if (generalAverage >= 96 && generalAverage <= 100) {
20        cout << "Applicable for Lance Sy Scholar." << endl;
21    }
22    else {
23        cout << "No Scholarship Available." << endl;
24    }
25
26    return 0;
27 }

```

Enter your General Average: 100
Applicable for Lance Sy Scholar.

=== Code Execution Successful ===

```

1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      float generalAverage;
6
7      cout << "Enter your General Average: ";
8      cin >> generalAverage;
9
10     if (generalAverage >= 81 && generalAverage <= 85) {
11         cout << "Applicable for SM Foundation" << endl;
12     }
13     else if (generalAverage >= 86 && generalAverage <= 90) {
14         cout << "Applicable for SMDC Scholar" << endl;
15     }
16     else if (generalAverage >= 91 && generalAverage <= 95) {
17         cout << "Applicable for NU Scholar" << endl;
18     }
19     else if (generalAverage >= 96 && generalAverage <= 100) {
20         cout << "Applicable for Lance Sy Scholar." << endl;
21     }
22     else {
23         cout << "No Scholarship Available." << endl;
24     }
25
26     return 0;
27 }

```

Enter your General Average: 79
No Scholarship Available.

=== Code Execution Successful ===

Finals

Comp Prog

April 24, 2025. Switch - Given code

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5
6     // Variable to be used as switch expression
7     char x = 'S';
8
9     // Switch statement with three cases
10    switch (x) {
11        case 'S':
12            cout << "S";
13            break;
14        case 'C':
15            cout << "C";
16            break;
17        default:
18            cout << "Other than S and C";
19            break;
20    }
21    return 0;
22 }
```

S

=== Code Execution Successful ===

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int day = 3;
6
7     switch (day) {
8         case 1:
9             cout << "Monday";
10            break;
11        case 2:
12            cout << "Tuesday";
13            break;
14        case 3:
15            cout << "Wednesday";
16            break;
17        case 4:
18            cout << "Thursday";
19            break;
20        case 5:
21            cout << "Friday";
22            break;
23        default:
24            cout << "Weekend";
25    }
26
27    return 0;
28 }
```

Wednesday

=== Code Execution Successful ===

```

1  # include <iostream>
2  using namespace std;
3
4  int main()
5  {
6      float num, average, sum = 0.0;
7      int i, n;
8
9      cout << "Maximum number of inputs: ";
10     cin >> n;
11
12     for(i = 1; i <= n; ++i)
13     {
14         cout << "Enter n" << i << ": ";
15         cin >> num;
16
17         if(num < 0.0)
18         {
19             // Control of the program move to jump:
20             goto jump;
21         }
22         sum += num;
23     }
24
25     jump:
26     average = sum / (i - 1);
27     cout << "\nAverage = " << average;
28     return 0;
29 }

```

Maximum number of inputs: 5

Enter n1: 5

Enter n2: 10

Enter n3: 15

Enter n4: 20

Enter n5: 25

Average = 15

=== Code Execution Successful ===

April 24, 2025. Activity 1 - Switch based on the following parameter: A - Ateneo, F- FEU, N - NU, D - DLSU, U - UST, S - Adamson, P - UP, E- UE

<pre>1 #include <iostream> 2 using namespace std; 3 4 int main() { 5 char schoolCode; 6 7 cout << "Select a university by entering the corresponding letter:\n"; 8 cout << "A - Ateneo\n"; 9 cout << "F - FEU\n"; 10 cout << "N - NU\n"; 11 cout << "D - DLSU\n"; 12 cout << "U - UST\n"; 13 cout << "S - Adamson\n"; 14 cout << "P - UP\n"; 15 cout << "E - UE\n"; 16 cout << "Enter your choice: "; 17 cin >> schoolCode; 18 19 // Convert input to uppercase to allow lowercase letters 20 schoolCode = toupper(schoolCode); 21 22 switch (schoolCode) { 23 case 'A': 24 cout << "You selected: Ateneo\n"; 25 break; 26 case 'F': 27 cout << "You selected: FEU\n"; 28 break; 29 case 'N': 30 cout << "You selected: NU\n"; 31 break; 32 case 'D': 33 cout << "You selected: DLSU\n"; 34 break; 35 case 'U': 36 cout << "You selected: UST\n"; 37 break; 38 case 'S': 39 cout << "You selected: Adamson\n"; 40 break; 41 case 'P': 42 cout << "You selected: UP\n"; 43 break; 44 case 'E': 45 cout << "You selected: UE\n"; 46 break; 47 default: 48 cout << "Invalid input. Please enter a valid letter from the list.\n"; 49 } 50 51 return 0; 52 }</pre>	<pre>Select a university by entering the corresponding letter: A - Ateneo F - FEU N - NU D - DLSU U - UST S - Adamson P - UP E - UE Enter your choice: A You selected: Ateneo === Code Execution Successful ===</pre>
<pre>1 #include <iostream> 2 using namespace std; 3 4 int main() { 5 char schoolCode; 6 7 cout << "Select a university by entering the corresponding letter:\n"; 8 cout << "A - Ateneo\n"; 9 cout << "F - FEU\n"; 10 cout << "N - NU\n"; 11 cout << "D - DLSU\n"; 12 cout << "U - UST\n"; 13 cout << "S - Adamson\n"; 14 cout << "P - UP\n"; 15 cout << "E - UE\n"; 16 cout << "Enter your choice: "; 17 cin >> schoolCode; 18 19 // Convert input to uppercase to allow lowercase letters 20 schoolCode = toupper(schoolCode); 21 22 switch (schoolCode) { 23 case 'A': 24 cout << "You selected: Ateneo\n"; 25 break; 26 case 'F': 27 cout << "You selected: FEU\n"; 28 break; 29 case 'N': 30 cout << "You selected: NU\n"; 31 break; 32 case 'D': 33 cout << "You selected: DLSU\n"; 34 break; 35 case 'U': 36 cout << "You selected: UST\n"; 37 break; 38 case 'S': 39 cout << "You selected: Adamson\n"; 40 break; 41 case 'P': 42 cout << "You selected: UP\n"; 43 break; 44 case 'E': 45 cout << "You selected: UE\n"; 46 break; 47 default: 48 cout << "Invalid input. Please enter a valid letter from the list.\n"; 49 } 50 51 return 0; 52 }</pre>	<pre>Select a university by entering the corresponding letter: A - Ateneo F - FEU N - NU D - DLSU U - UST S - Adamson P - UP E - UE Enter your choice: B Invalid input. Please enter a valid letter from the list. === Code Execution Successful ===</pre>

April 25, 2025. Hands-On Quiz - Switch that will display and select among the category: Shoes, Clothing, Make-up, and Bags

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     char category;
6
7     cout << "Enter a corresponding letter from A to D:\n";
8     cout << "Enter your choice: ";
9     cin >> category;
10
11     switch (category) {
12         case 'A':
13             cout << "You selected: Shoes\n";
14             break;
15         case 'B':
16             cout << "You selected: Clothing\n";
17             break;
18         case 'C':
19             cout << "You selected: Make-up\n";
20             break;
21         case 'D':
22             cout << "You selected: Bags\n";
23             break;
24         default:
25             cout << "Invalid input. Please enter a valid letter from the list.\n";
26     }
27
28     return 0;
29 }
```

Enter a corresponding letter from A to D:
Enter your choice: C
You selected: Make-up

=== Code Execution Successful ===

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     char category;
6
7     cout << "Enter a corresponding letter from A to D:\n";
8     cout << "Enter your choice: ";
9     cin >> category;
10
11     switch (category) {
12         case 'A':
13             cout << "You selected: Shoes\n";
14             break;
15         case 'B':
16             cout << "You selected: Clothing\n";
17             break;
18         case 'C':
19             cout << "You selected: Make-up\n";
20             break;
21         case 'D':
22             cout << "You selected: Bags\n";
23             break;
24         default:
25             cout << "Invalid input. Please enter a valid letter from the list.\n";
26     }
27
28     return 0;
29 }
```

Enter a corresponding letter from A to D:
Enter your choice: F
Invalid input. Please enter a valid letter from the list.

=== Code Execution Successful ===

April 30, 2025. Given code

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int scores[10];
6     int i;
7     int max, min;
8     float sum = 0, average;
9
10    // Input scores
11    cout << "Enter the scores of 10 students:\n";
12    for (i = 0; i < 10; i++) {
13        cout << "Score #" << i + 1 << ": ";
14        cin >> scores[i];
15        sum += scores[i];
16    }
17
18    // Initialize max and min
19    max = scores[0];
20    min = scores[0];
21
22    // Find max and min
23    for (i = 1; i < 10; i++) {
24        if (scores[i] > max) {
25            max = scores[i];
26        }
27        if (scores[i] < min) {
28            min = scores[i];
29        }
30    }
31
32    // Calculate average
33    average = sum / 10;
34
35    // Display results
36    cout << "\nScores entered: ";
37    for (i = 0; i < 10; i++) {
38        cout << scores[i] << " ";
39    }
40
41    cout << "\nHighest Score: " << max;
42    cout << "\nLowest Score: " << min;
43    cout << "\nAverage Score: " << average << endl;
44
45    return 0;
46 }
```

Enter the scores of 10 students:
Score #1: 8
Score #2: 4
Score #3: 5
Score #4: 6
Score #5: 9
Score #6: 7
Score #7: 5
Score #8: 1
Score #9: 3
Score #10: 5

Scores entered: 8 4 5 6 9 7 5 1 3 5
Highest Score: 9
Lowest Score: 1
Average Score: 5.3

=== Code Execution Successful ===

```
1 #include <stdio.h>
2
3 int main() {
4     int numbers[5] = {10, 20, 30, 40, 50};
5     int s;
6
7     printf ("The array elements are:\n");
8     for(s = 0; s < 5; s++) {
9         printf ("%d\n", numbers[s]);
10    }
11
12    return 0;
13 }
```

The array elements are:
10
20
30
40
50

=== Code Execution Successful ===

<pre>1 #include <stdio.h> 2 3 int main() { 4 int numbers[2][3] = { 5 {10, 20, 30}, 6 {40, 50, 60} 7 }; 8 9 int s, x; 10 11 printf("The 2D array elements are:\n"); 12 for(s = 0; s < 2; s++) { 13 for(x = 0; x < 3; x++) { 14 printf("%d ", numbers[s][x]); 15 } 16 printf("\n"); 17 } 18 19 return 0; 20 } 21 }</pre>	The 2D array elements are: 10 20 30 40 50 60 === Code Execution Successful ===
--	--

April 30, 2025. Given string code

<pre>1 #include <stdio.h> 2 3 int main() { 4 5 const char *schools[] = { 6 "UP", 7 "UST", 8 "DLSU", 9 "ADMU", 10 "ADU", 11 "FEU", 12 "UE", 13 "NU" 14 }; 15 16 int x; 17 printf("List of schools:\n"); 18 for(x = 0; x < 7; x++) { 19 printf("%s\n", schools[x]); 20 } 21 22 return 0; 23 }</pre>	List of schools: UP UST DLSU ADMU ADU FEU UE === Code Execution Successful ===
---	--

May 02, 2025. Final Act 1 - Using array & string using months of the year

- a. Display all at once
- b. Asking user input of 1 to 12

<pre> 1 #include <iostream> 2 using namespace std; 3 4 int main() { 5 string months[] = { 6 "January", "February", "March", "April", 7 "May", "June", "July", "August", 8 "September", "October", "November", "December" 9 }; 10 11 // Display all months 12 for (int i = 0; i < 12; i++) { 13 cout << months[i] << endl; 14 } 15 16 // Ask for user input 17 int num; 18 cout << "\nEnter a number (1-12): "; 19 cin >> num; 20 21 // Display the selected month 22 if (num >= 1 && num <= 12) { 23 cout << "You selected: " << months[num - 1] << endl; 24 } else { 25 cout << "Invalid input!" << endl; 26 } 27 28 return 0; 29 } </pre>	<pre> January February March April May June July August September October November December Enter a number (1-12): 8 You selected: August === Code Execution Successful === </pre>
--	--

May 09, 2025. Given code

<pre> 1 #include <iostream> 2 using namespace std; 3 4 int main() { 5 const int MAX = 100; 6 string names[MAX]; 7 int marks[MAX]; 8 char grades[MAX]; 9 int numStudents; 10 int totalMarks = 0; 11 12 cout << "Enter number of students: "; 13 cin >> numStudents; 14 15 // Input names and marks 16 for (int i = 0; i < numStudents; i++) { 17 cout << "\nEnter name of student " << i + 1 << ": "; 18 cin >> names[i]; 19 20 cout << "Enter marks (0-100): "; 21 cin >> marks[i]; 22 23 // Assign grade 24 if (marks[i] >= 90) 25 grades[i] = 'A'; 26 else if (marks[i] >= 80) 27 grades[i] = 'B'; 28 else if (marks[i] >= 70) 29 grades[i] = 'C'; 30 else if (marks[i] >= 60) 31 grades[i] = 'D'; 32 else if (marks[i] >= 50) 33 grades[i] = 'E'; 34 else 35 grades[i] = 'F'; 36 37 totalMarks += marks[i]; 38 } 39 40 // Output student results 41 cout << "\nStudent Results:\n"; 42 for (int i = 0; i < numStudents; i++) { 43 cout << names[i] << " - Marks: " << marks[i] << ", Grade: " << grades[i] << endl; 44 } 45 46 // Class average 47 float average = (float) totalMarks / numStudents; 48 cout << "\nClass Average: " << average << endl; 49 50 return 0; 51 } </pre>	<pre> Enter number of students: 5 Enter name of student 1: Alvey Enter marks (0-100): 90 Enter name of student 2: Denzul Enter marks (0-100): 92 Enter name of student 3: Carl Enter marks (0-100): 98 Enter name of student 4: Joe Enter marks (0-100): 87 Enter name of student 5: Drey Enter marks (0-100): 80 Student Results: Alvey - Marks: 90, Grade: A Denzul - Marks: 92, Grade: A Carl - Marks: 98, Grade: A Joe - Marks: 87, Grade: B Drey - Marks: 80, Grade: B Class Average: 89.4 === Code Execution Successful === </pre>
--	--

May 14, 2025. Given code to debug

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 // Month class definition
6 class Month {
7 private:
8     int monthNumber;
9     string monthName;
10
11 public:
12     // Constructor
13     Month(int number = 1, string name = "January") {
14         monthNumber = number;
15         monthName = name;
16     }
17
18     // Setters
19     void setMonthNumber(int number) {
20         monthNumber = number;
21     }
22
23     void setMonthName(string name) {
24         monthName = name;
25     }
26
27     // Getters
28     int getMonthNumber() const {
29         return monthNumber;
30     }
31
32     string getMonthName() const {
33         return monthName;
34     }
35
36     // Display function
37     void displayMonth() const {
38         cout << "Month " << monthNumber << ": " << monthName << endl;
39     }
40 };
41
42 // Main function to demonstrate usage
43 int main()
44 {
45     // Array of month names
46     string monthName[12] = {
47         "January", "February", "March", "April",
48         "May", "June", "July", "August",
49         "September", "October", "November", "December"
50     };
51
52     // Error message
53     ERROR!
54     /tmp/58wOX8g2y/main.cpp:45:5: error: expected initializer before 'string'
55     45 |     string monthName[12] = {
56         |         ~~~~~
57
58     === Code Exited With Errors ===
```

Fixed code

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 // Month class definition
6 class Month {
7 private:
8     int monthNumber;
9     string monthName;
10
11 public:
12     // Constructor
13     Month(int number = 1, string name = "January") {
14         monthNumber = number;
15         monthName = name;
16     }
17
18     // Setters
19     void setMonthNumber(int number) {
20         monthNumber = number;
21     }
22
23     void setMonthName(string name) {
24         monthName = name;
25     }
26
27     // Getters
28     int getMonthNumber() const {
29         return monthNumber;
30     }
31
32     string getMonthName() const {
33         return monthName;
34     }
35
36     // Display function
37     void displayMonth() const {
38         cout << "Month " << monthNumber << ": " << monthName << endl;
39     }
40 };
41
42 // Main function to demonstrate usage
43 int main() {
44     // Array of month names
45     string monthName[12] = {
46         "January", "February", "March", "April",
47         "May", "June", "July", "August",
48         "September", "October", "November", "December"
49     };
50
51     // Loop to create and display all 12 months
52     for (int i = 0; i < 12; ++i) {
53         Month m(i + 1, monthName[i]);
54         m.displayMonth();
55     }
56
57     return 0;
58 }
```

Month 1: January
Month 2: February
Month 3: March
Month 4: April
Month 5: May
Month 6: June
Month 7: July
Month 8: August
Month 9: September
Month 10: October
Month 11: November
Month 12: December

=== Code Execution Successful ===

May 23, 2025. Finals Exam - flowchart on paper and code to be written in notepad

May 23, 2025. Finals Exam Question 1 - Sum and Average Calculator

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      double num1, num2, num3, sum, average;
6
7      cout << "Enter first number" << endl;
8      cin >> num1;
9
10     cout << "Enter second number" << endl;
11     cin >> num2;
12
13     cout << "Enter third number" << endl;
14     cin >> num3;
15
16     sum = num1 + num2 + num3;
17     average = sum / 3;
18
19     cout << "Sum: " << sum << endl;
20     cout << "Average: " << average << endl;
21
22     return 0;
23
24 }
```

```
Enter first number
3
Enter second number
7
Enter third number
9
Sum: 19
Average: 6.33333
```

=== Code Execution Successful ===

May 23, 2025. Finals Exam Question 2 - Even or Odd Checker using if statement

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int number;
6
7     cout << "Enter an integer" << endl;
8     cin >> number;
9
10    if (number % 2 == 0) {
11        cout << number << " is even" << endl;
12    } else {
13        cout << number << " is odd" << endl;
14    }
15
16    return 0;
17
18 }
```

Enter an integer
7
7 is odd

=== Code Execution Successful ===

```
1 #include <iostream>
2 using namespace std;
3
4 int main() {
5     int number;
6
7     cout << "Enter an integer" << endl;
8     cin >> number;
9
10    if (number % 2 == 0) {
11        cout << number << " is even" << endl;
12    } else {
13        cout << number << " is odd" << endl;
14    }
15
16    return 0;
17
18 }
```

Enter an integer
12
12 is even

=== Code Execution Successful ===

May 23, 2025. Finals Exam Question 3 - Find the Largest of Three Numbers, nested if-else

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int num1, num2, num3;
6
7      cout << "Enter first number" << endl;
8      cin >> num1;
9
10     cout << "Enter second number" << endl;
11     cin >> num2;
12
13     cout << "Enter third number" << endl;
14     cin >> num3;
15
16     if (num1 > num2) {
17         if (num1 > num3) {
18             cout << "The largest number is: " << num1 << endl;
19         } else {
20             cout << "The largest number is: " << num3 << endl;
21         }
22     } else {
23         if (num2 > num3) {
24             cout << "The largest number is: " << num2 << endl;
25         } else {
26             cout << "The largest number is: " << num3 << endl;
27         }
28     }
29
30     return 0;
31 }
```

Enter first number
97
Enter second number
85
Enter third number
88
The largest number is: 97

=== Code Execution Successful ===